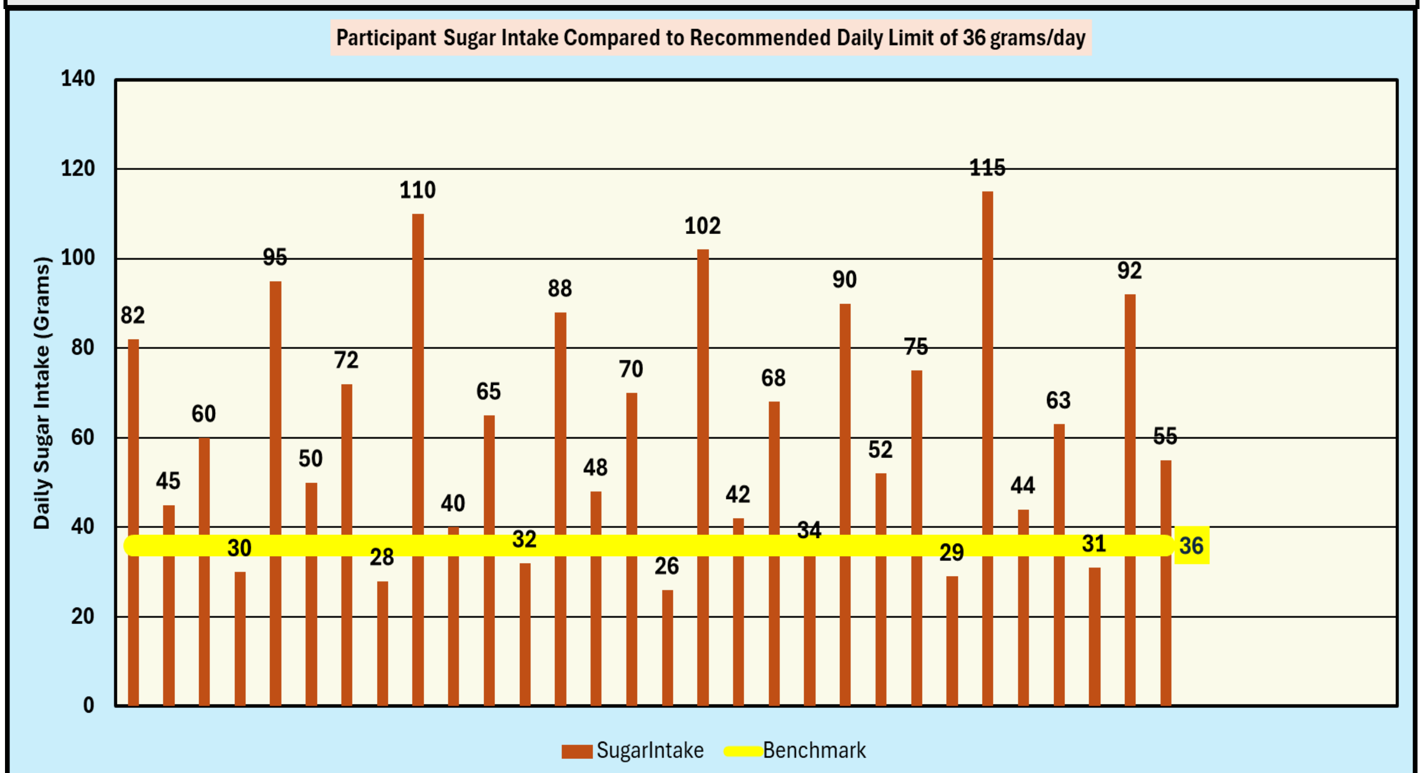
Project title:	**Wellness and Nutrition Behaviors Across Age Groups**
Your name:	Krisine Steele
Project date:	2/24/2026
Describe your data domain:	This project explores health-related behaviors, including nutrition intake, hydration, sleep duration, exercise, and age-group patterns. The data includes both healthy and unhealthy food categories, lifestyle behaviors such as water intake, sleep hours, and exercise minutes, and demographic information.
Describe your data source:	This dataset was generated using Microsoft Copilot 365's data-creation tools. It is an AI-generated synthetic dataset designed for analysis and visualization practice within this project.
Clickable link to data source:	Because the dataset was created in-app, there is no public link. It originates from Microsoft Copilot 365's internal data generation feature.
Number of data columns:	26
Number of data records:	30
Summarize each data feature/column in a table:	ID, AgeGroup, SugarIntake, ExerciseMinutes, CoffeeCups, TeaCups, WaterIntake, SleepHours, ProcessedFood, FastFood, Soda, WholeGrains, Dairy, VinegarOil, Avocados, Fish, Chicken, Fruit, Apples, Eggs, Candy, CakePastries, BeansLegumes, Salad, Veggies, Benchmark (of sugar=36)
Describe any assumptions:	*Age groups are assumed to represent distinct and comparable adult life stages. *Servings are assumed to be measured consistently for all individuals. *Sleep ≥ 7 hours is considered meeting the recommended amount for adults (CDC guideline). *Water intake ≥ 8 cups/day is considered meeting hydration recommendations. *Junk-food intake is measured using a composite score (processed foods, fast food, soda, candy, and pastries). *All records are assumed to be accurately generated and internally consistent. *Benchmark column is assumed to be a fixed reference value not needed for analysis.
Goal 1:	**Sugar Intake Benchmark**-Determine how the group's average daily sugar intake compares to the recommended daily limit of 36 grams to assess whether the class is meeting or exceeding a widely accepted health guideline. A benchmark chart clearly compares a single measured value to a recommended standard, making it easy to see whether the group's average sugar intake is within a healthy range.
Goal 2:	**Average Daily Exercise Minutes by Age Group**-Compare average daily exercise minutes across age groups to understand how physical activity levels differ at various life stages. A bar or histogram-style chart allows for easy comparison of average values across categories, making differences in exercise levels between age groups clear and direct.
Goal 3:	**Healthy Vs. Unhealthy Food Choices**-Examine how much of the group's total food intake comes from healthy foods versus unhealthy foods to understand the balance of eating habits. A treemap effectively displays parts of a whole, showing the proportion of healthy vs. unhealthy foods and highlighting which categories contribute most within each group.
Goal 4:	**Wellness Behaviors-**Show how participants distribute across healthy vs. below-benchmark sleep, hydration, and junk-food intake categories using a single hierarchical visual. A sunburst chart organizes multiple related wellness behaviors into clear hierarchical rings, making it easy to see how each domain contributes to overall wellness.
Goal 5:	**Healhty Food Rankings by Age Group**-Compare how different age groups rank various healthy food categories to understand differences in dietary preferences across age. A bump chart highlights changes in ranking across categories, making it easy to compare how each age group prioritizes different types of healthy foods.

Goal 1:	Sugar Intake Benchmark
Goal:	Compare the group's average daily sugar intake to the recommended daily limit to evaluate how participants measure up to a widely accepted health guideline.
Single Attribute:	Sugar Intake (grams/day)
Attribute Summary as a Table:	

Metric	Value
Average Sugar Intake (g/day)	=AVERAGE(SugarIntake column)
Recommended Intake (g/day)	36 grams

Selected Chart Type:	Benchmark Chart
Why is this a good type for this goal:	A benchmark chart is ideal for comparing a single measured value to a recommended standard. Sugar intake is a widely recognized health indicator, and this chart clearly shows whether the group's average intake falls within or exceeds healthy guidelines. This makes the overall health assessment simple, visual, and immediately interpretable.



Data Story: TOO MUCH SUGAR

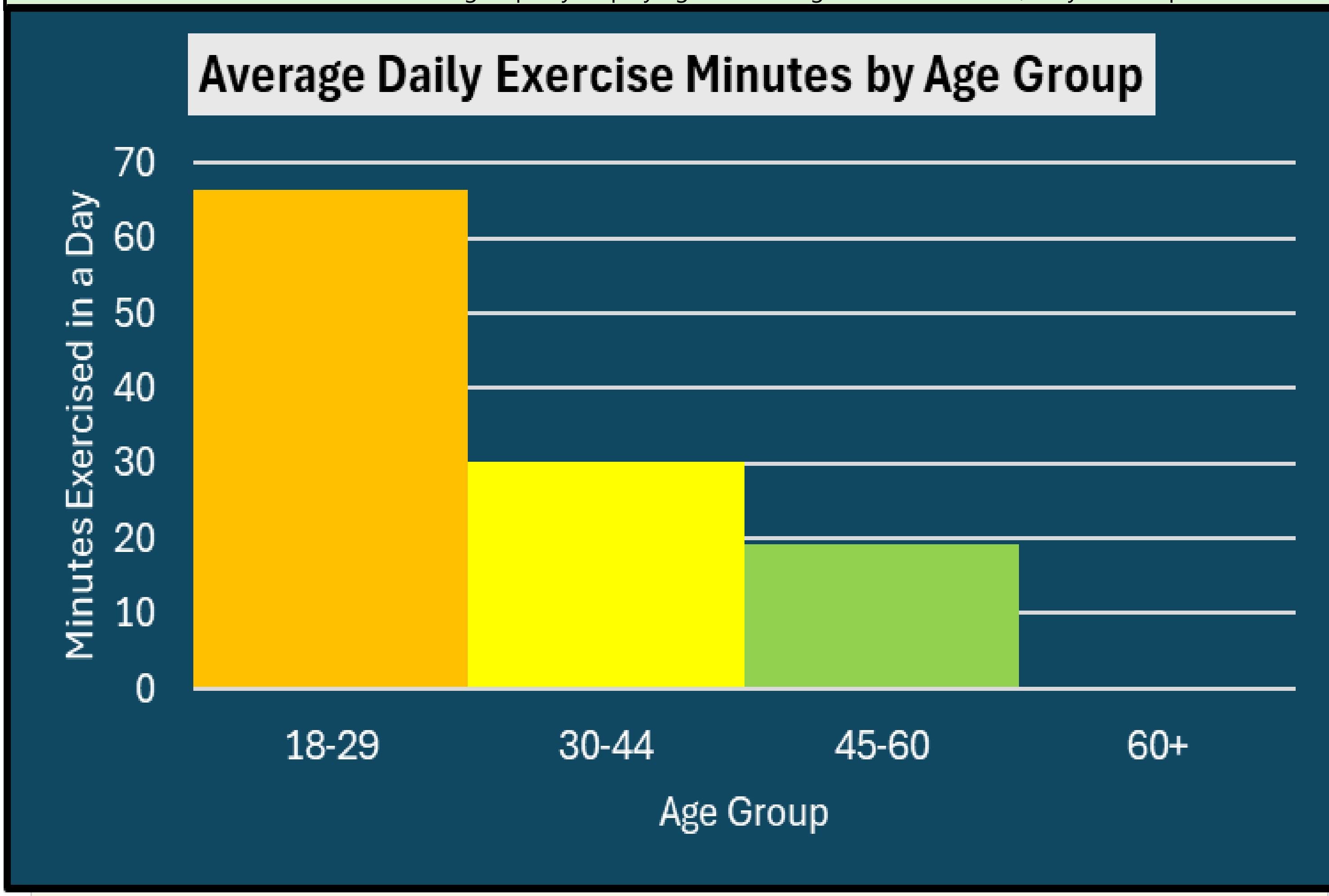
In this sample of 30 participants, the average daily sugar intake was **61.1 grams**, which is **25.1 grams above** the 36-gram recommended limit. **About 77%** of participants exceeded the benchmark, and among those who were over, the average excess was **34.6 grams per day**. This is nearly double the recommended limit! Only a small number fell below the recommended limit, highlighting a consistent pattern of elevated sugar intake across the group.

Goal 1:	Average Daily Exercise Minutes by Age Group
Goal:	Determine how much participants in each age group exercise on a typical day.
Multiple Attributes:	Age Group, Exercise Minutes

Attribute Summary as a Table:

Age Group	Average Exercise Minutes
18–29	66.43
30–44	30.22
45–60	19.33

Selected Chart Type:	Histogram Chart
Why is this a good type for this goal:	This histogram-style chart clearly shows how exercise levels decline across age groups by displaying their averages in a continuous, easy-to-compare format.



Data Story-Energy Levels for Exercise Decline Over Time

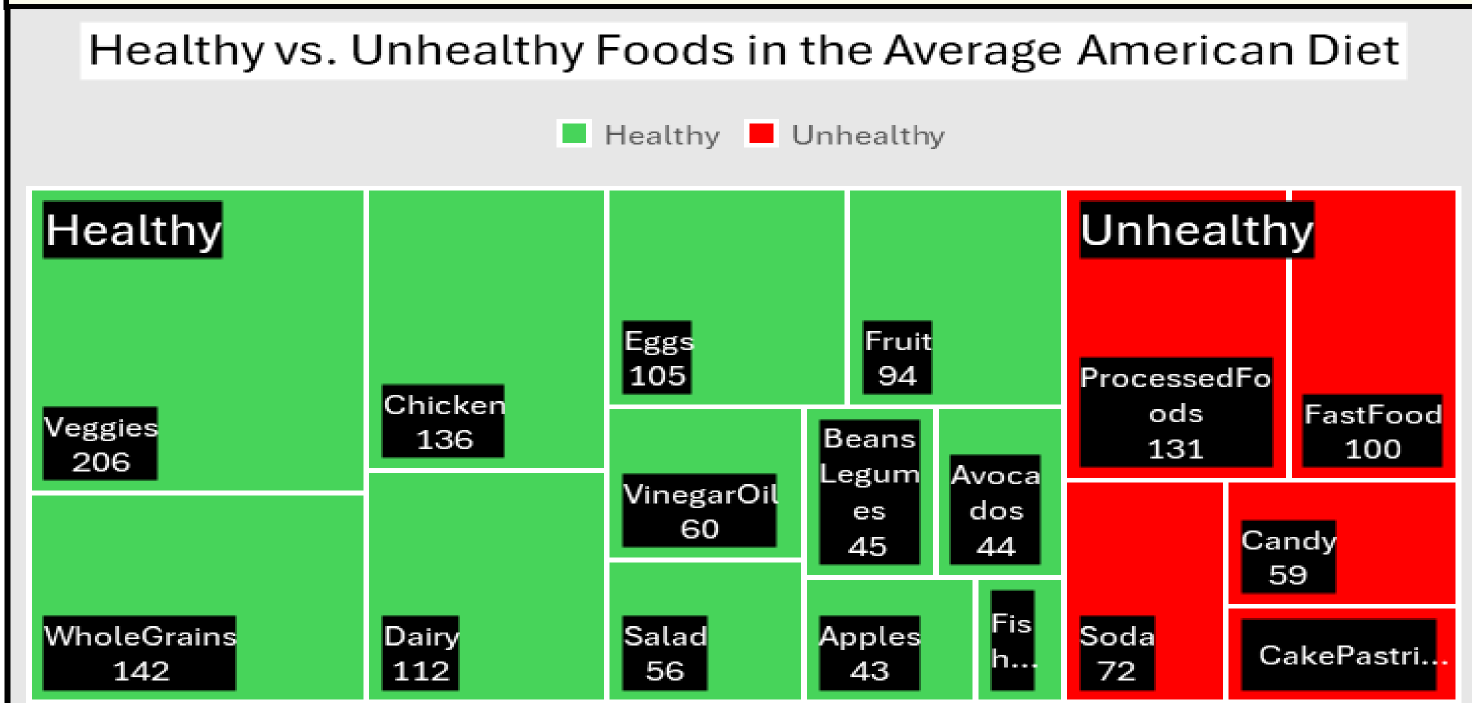
The data clearly shows that younger participants exercise far more than older ones. In this sample, the 18–29 age group averages *66.43 minutes* of exercise per day, the highest of all age ranges. Once participants reach their 30s, average daily exercise drops sharply to *30.22 minutes*, which is less than half the activity of the youngest group. This decline continues into the 45–60 age range, where the average falls to *19.33 minutes per day*. By age 60 and older, participants report *0 minutes* of daily exercise. Overall, the pattern highlights a steady loss of daily physical activity as age increases, suggesting meaningful differences in energy, health, or lifestyle across the lifespan.

Goal 3:	Healthy Versus Unhealthy Food Choices
Goal:	Summarize how much of the group's total food intake comes from **healthy foods** versus **unhealthy foods** to understand the overall balance of eating habits.
Multiple Attributes:	Healthy or Unhealthy

Attribute Summary as a Table:

Group	Items Included	Total Amount
Healthy Foods	WholeGrains, Dairy, VinegarOil, Avocados, Fish, Chicken, Fruit, Apples, Eggs, BeansLegumes, Salad, Veggies	*1,065*
Unhealthy Foods	ProcessedFoods, FastFood, Soda, Candy, CakesPastries	*407*

Selected Chart Type:	Tree Map
Why is this a good type for this goal:	A treemap is ideal for showing parts of a whole. It clearly displays the overall share of healthy versus unhealthy foods, while also showing which specific foods contribute most within each group.



Data Story-Room for Improvementin our Unhealthy Choices?

The data shows that **72.35%** of total intake (1,065 points) comes from **healthy foods**, while **27.65%** (407 points) comes from **unhealthy foods**. Healthy items include **vegetables, whole grains, dairy, chicken, fruit, eggs, beans/legumes, salad, fish, avocados, and vinegar/oil**, with **vegetables (206)**, **whole grains (142)**, and **chicken (136)** together making up **nearly one-third** of the overall diet. Unhealthy foods—**processed foods, fast food, soda, candy, and cakes/pastries—**are largely driven by **processed foods (131)** and **fast food (100)**, accounting for **over half** of all unhealthy intake. Overall, the treemap shows that healthy foods dominate the diet, but a small set of high-impact unhealthy items still contributes a notable share.

Goal 4: Wellness Behaviors:Sleep, Hydration and Diet

Goal: Show how our class of 30 distributes across healthy vs. below benchmark sleep and hydration, and across high/moderate/low junk-food intake using a single hierarchal view.

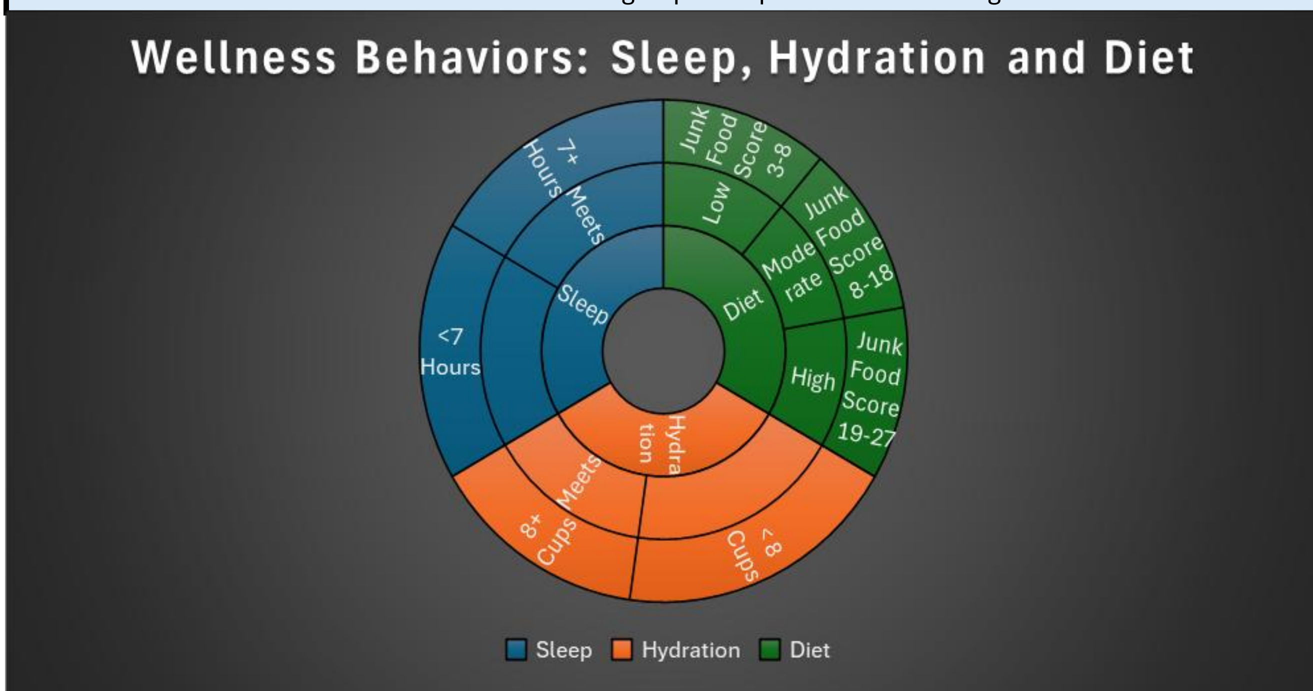
Multiple Attributes: Sleep, hydration and diet behavior

Attribute Summary as a Table:

Domain	Bucket	Count	% of Class
Sleep	Meets (7+ hours)	15	50%
Sleep	Below (<7 hours)	15	50%
Hydration	Meets (8+ cups)	13	43%
Hydration	Below (<8 cups)	17	57%
Diet	Low (junk-food intake 3-8)	10	33%
Diet	Medium (junk-food intake 8-18)	10	33%
Diet	High (junk-food intake 19-27)	10	33%

Selected Chart Type: Sunburst Chart

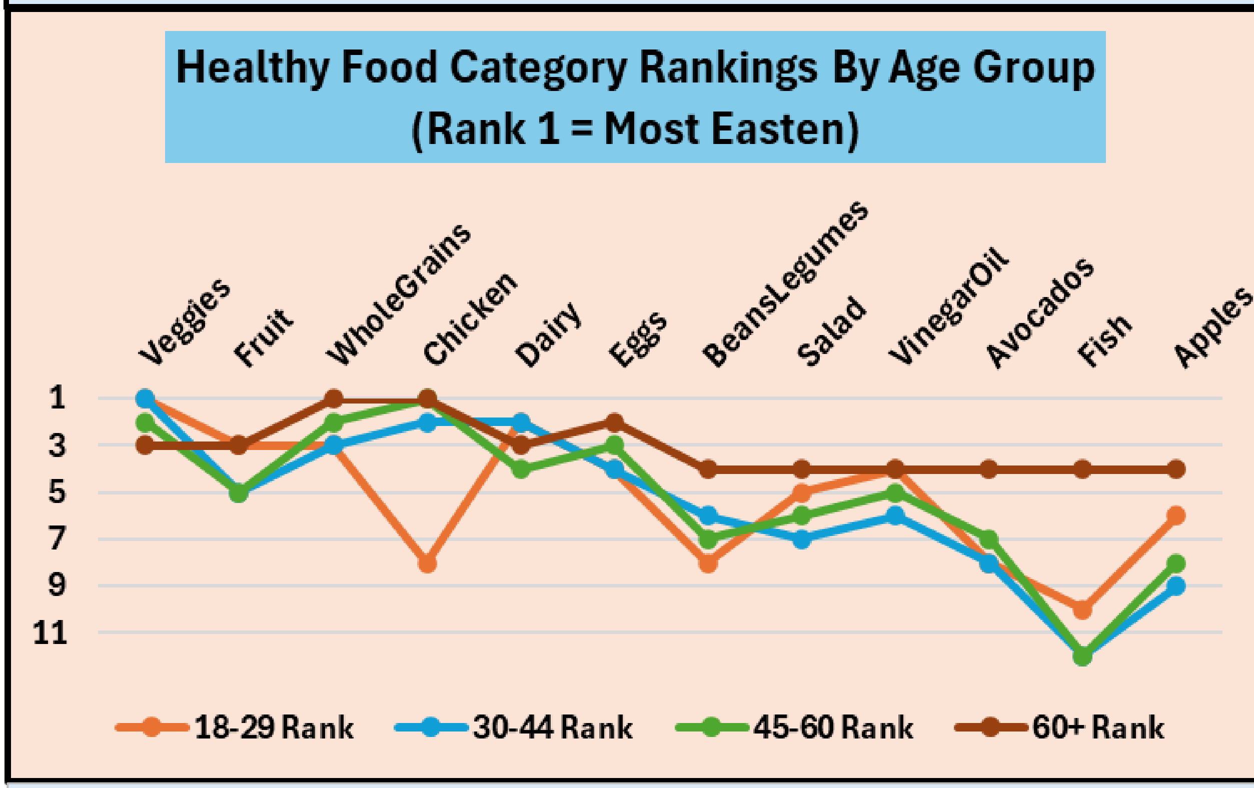
Why is this a good type for this goal: A sunburst chart displays hierachal categories as parts of a whole. It clearly shows how Sleep, Hydration and Diet contribute to overall wellness and how subgroups compare within each ring.



Data Story-Wellness Behaviors

In a class of 30, sleep is split evenly (**15 meet 7+ hours; 15 below**) hydration trails (**13 meet 8+ cups; 17 below**), and a composite junk-food score places students evenly into **Low, Moderate, and High** groups (**10 each; High = 19–27**). The Sunburst chart highlights some serious opportunities for health improvement across the general population. Hydration has the largest gap with 57% of people not drinking enough water. One-third of the population has a high junk-food intake, and there's a 50/50 sleep split—clear priorities for improvement**. Sleep more, drink more water and eat less junk food.**

Goal 5:	Healthy Food Category Rankings by Age Group
Goal:	To compare how different age groups rank various healthy food categories, identifying which foods are most and least preferred within each age group.
Multiple Attributes:	Healthy Categories, Age Ranges and Rankings
Attribute Summary as a Table:	
Healthy Categories	Age Group and Rankings
Veggies, Fruits, WholeGrains, Chicken, Dairy, Eggs, BeansLegumes, Salad, VinegarOil, Avocados, Fish, Apples	Age Groups: 18-29, 30-44, 45-60, 60+ Rankings: 1-12
Selected Chart Type:	Bump Chart (Line Chart with Ranks)
Why is this a good type for this goal:	A bump chart highlights how the ranking of categories changes across groups, making it easy to see which foods are ranked highest or lowest within each age group. This visual emphasizes shifts in preference and clearly displays how each age group's healthy-food choices differ from one another.



Data Story- Who Likes What Healthy Foods When?

This bump chart ranks healthy food categories within each age group. Among 18-29, the top choices are veggies (86) and dairy (42), while chicken ranks much lower. The 30-44 group favors veggies (63) and ranks chicken (41) and dairy (39) close behind, with fish at their lowest. In contrast, the 45-60 group ranks chicken highest (62) and places whole grains and veggies next (55 each), showing a shift towards protein and grains. The 60+ group shows limited variety, with small but highest total in whole grains and chicken (12 each). Overall, younger adults lean toward produce, mid-adults balance produce and protein, and older adults report narrower intake.

ID	AgeGrou	SugarIn	Exerci	Coffe	TeaCu	Water	SleepH	Proce	FastF	Soda	Whol	Dairy	Vineg	Avoca	Fish	Chick	Fruit	Apple	Eggs	Canc	Cakes	Beans	Salac	Veggi	Ber
1	30-44	82	10	2	0	5	6.5	6	3	4	1	4	1	0	0	5	1	0	3	3	2	1	0	3	36
2	45-60	45	40	1	1	7	7.5	3	1	1	4	5	3	2	1	4	3	1	4	1	1	2	2	7	36
3	30-44	60	20	2	0	6	6	5	2	3	2	4	1	1	0	6	2	1	3	2	2	1	1	5	36
4	18-29	30	55	0	2	9	8	2	1	0	5	6	4	3	2	3	5	2	5	0	0	3	3	10	36
5	45-60	95	5	3	0	4	6	7	7	4	5	1	0	0	0	6	1	0	2	4	3	0	0	2	36
6	30-44	50	35	1	1	8	7	3	2	2	5	5	3	2	1	4	4	2	4	1	1	2	2	8	36
7	45-60	72	15	2	0	5	6	6	5	3	4	2	1	1	0	5	2	1	3	3	2	1	1	4	36
8	18-29	28	60	0	2	10	8.5	2	1	1	6	6	4	3	2	3	6	3	5	0	0	3	4	12	36
9	45-60	110	0	3	0	4	6	7	7	5	6	1	0	0	0	6	1	0	2	5	3	0	0	1	36
10	30-44	40	45	1	1	8	7	3	2	1	5	5	3	2	1	4	4	2	4	1	1	2	3	9	36
11	45-60	65	25	2	0	6	6.5	5	3	3	3	3	1	1	0	5	2	1	3	2	2	1	1	6	36
12	18-29	32	70	0	2	10	8	2	1	1	6	6	4	3	2	3	6	3	5	0	0	3	4	13	36
13	45-60	88	5	3	0	4	6	7	7	4	5	1	0	0	0	6	1	0	2	4	3	0	0	2	36
14	30-44	48	38	1	1	7	7	3	2	2	5	5	3	2	1	4	4	2	4	1	1	2	3	8	36
15	45-60	70	18	2	0	5	6	6	5	3	4	2	1	1	0	5	2	1	3	3	2	1	1	5	36
16	18-29	26	75	0	2	11	8.5	2	1	1	6	6	4	3	2	3	6	3	5	0	0	3	4	14	36
17	60+	102	0	3	0	4	6	7	7	5	6	1	0	0	0	6	1	0	2	5	3	0	0	1	36
18	45-60	42	50	1	1	8	7.5	3	2	1	5	5	3	2	1	4	4	2	4	1	1	2	3	10	36
19	30-44	68	22	2	0	6	6.5	5	3	3	3	3	1	1	0	5	2	1	3	2	2	1	1	6	36
20	18-29	34	65	0	2	10	8	2	1	1	6	6	4	3	2	3	6	3	5	0	0	3	4	12	36
21	45-60	90	8	3	0	4	6	7	7	4	5	1	0	0	0	6	1	0	2	4	3	0	0	2	36
22	30-44	52	32	1	1	8	7	3	2	1	5	5	3	2	1	4	4	2	4	1	1	2	3	9	36
23	45-60	75	12	2	0	5	6	6	5	3	4	2	1	1	0	5	2	1	3	3	2	1	1	4	36
24	18-29	29	72	0	2	11	8.5	2	1	1	6	6	4	3	2	3	6	3	5	0	0	3	4	13	36
25	60+	115	0	3	0	4	6	7	7	5	6	1	0	0	0	6	1	0	2	5	3	0	0	1	36
26	45-60	44	48	1	1	8	7	3	2	1	5	5	3	2	1	4	4	2	4	1	1	2	3	10	36
27	30-44	63	28	2	0	6	6.5	5	3	3	3	3	1	1	0	5	2	2	3	2	2	1	1	6	36
28	18-29	31	68	0	2	10	8	2	1	1	6	6	4	3	2	3	6	3	5	0	0	3	4	12	36
29	45-60	92	6	3	0	4	6	7	7	4	5	1	0	0	0	6	1	0	2	4	3	0	0	2	36
30	30-44	55	42	1	1	8	7	3	2	1	5	5	3	2	1	4	4	2	4	1	1	2	3	9	36